# The Whole Shabang

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Ladder Explained

Now this is the right place to slow down and really understand what the ladder test actually is, why it exists, and how to read it without fooling yourself. I'll go deep, but keep it practical and grounded in real-world shooting, not internet mythology.

What the Ladder Test Really Is (and Why It Exists)

At its core, a ladder test is not about group size. A ladder test is a pressure-vs-velocity-vs-barrel timing diagnostic.

It exists to show you where small increases in powder stop moving point of impact vertically. That's it.

Everything else people claim it does is secondary. The Physics Behind the Ladder (why it works) When you increase powder charge:

Velocity increases

Barrel time changes

Barrel whip timing changes

Bullet exits the muzzle at a different phase of barrel vibration

This causes:

Vertical POI shift, even with perfect shooting

Key concept: barrel harmonics

Every barrel vibrates like a tuning fork when fired. There are moments in the vibration cycle where:

The muzzle is moving very little vertically

Bullet exit timing is more forgiving

These moments create "nodes".

A ladder test helps you find charge weights where velocity increases, but vertical POI does NOT. That is the magic.

What a Ladder Test Is NOT

Let's kill some myths:

■ It is NOT for finding the smallest group

■ It is NOT a replacement for OCW

■ It is NOT about SD

■ It is NOT reliable at 100 yd (unless you know what you're doing)

A ladder is about vertical trends, not clusters.

Why Distance Matters (a lot) - Vertical dispersion grows with distance.

At 100 m / yd → velocity differences hide

At 300+ m / yd → nodes become obvious

At 600 m / yd → they scream at you

Ideal ladder distances:

300 yd minimum

400–600 yd preferred

This is why ladder tests were originally a long-range benchrest tool, not a 100-m / yd range trick.

How a Proper Ladder Test Is Shot

Load prep:

Same brass lot

Same primer

Same seating depth

Charges usually in 0.2–0.3 gr increments

One round per charge

Example:

41.0, 41.3, 41.6,41.9,42.2,42.5

Shooting method:

One aiming point

Shoot low to high charge

Let barrel cool between shots

Same wind condition if possible

Each shot:

Different charge

Different velocity

Different barrel timing

How to Read Ladder Results (this is the heart of it). After shooting, ignore horizontal dispersion.

You care about vertical only.

You'll typically see:

Shots climbing steadily

Then two or more shots land at nearly the same vertical height

Then climb resumes

Example (vertical only):

42.5 | o

42.2 | o

41.9 | o o <-- NODE

41.6 | o o <-- NODE

41.3 | o

41.0 | o

That flat spot = Likely a node

What it means:

Velocity increased

Barrel timing stayed favorable

Bullet exited at same muzzle position

That charge region is inherently forgiving.

Why Flat Vertical = Forgiveness

Let’s say:

You load 41.9 today.

Next month it’s hotter

Pressure rises

Velocity increases slightly

If you’re in a node:

POI barely changes

If you’re between nodes:

POI shifts dramatically

That’s why ladder-found nodes: Hold vertical at distance, Track temperature better

Don’t “fall apart” in matches

Why One Round Per Charge Is Enough

This confuses people.

You’re not averaging anything. You’re not grouping. You’re mapping a trend. Think of it like plotting a graph:

X-axis = charge and Y-axis = vertical POI

You don’t need repeats to see slope changes. Pressure Signs During the Ladder. This is also where ladders shine

As you climb charges:

Bolt lift

Primer flattening

Ejector marks

Velocity jumps

If pressure spikes:

Stop immediately

Node above it is irrelevant

You’ve just defined your safe ceiling.

Ladder vs OCW (why ladder comes first)

Ladder test answers:

“Where are the barrel timing nodes and safe pressure limits?”

OCW answers:

“Which of these nodes is the most repeatable in group form?”

You use ladder results to:

Narrow powder range

Eliminate bad regions

Avoid wasting OCW components

Common Ladder Mistakes (and why people say they “don’t work”)

Shooting at 100 yd

Shooting multiple shots per charge

Chasing group size

Shooting too fast (heat shifts POI)

Ignoring wind vertical

Using inconsistent brass

When a Ladder Test Is Most Useful

✔ New powder

✔ New bullet weight

✔ Unknown pressure window

✔ Long-range rifle

✔ Limited components

If you already know the system well, OCW alone can be enough — but ladder gives you context and safety.

How Ladder Results Feed the OCW

After the ladder:

You’ll usually see 1–2 viable nodes in green in the App. Pick one (often lower for safety)

Run OCW inside that node

Example:

Ladder node at 41.6–42.0

OCW charges: 41.6 / 41.8 / 42.0

Now OCW has a much higher success rate.

Final Mental Model (this is the key takeaway)

Ladder = maps barrel behaviour

OCW = validates stability

Seating depth = sharpens precision

SD = consistency, not accuracy

One-sentence ladder summary:

A ladder test reveals charge weights where increasing velocity stops moving vertical POI, identifying forgiving barrel timing nodes before you ever worry about group size.

OCW Explained I’ll walk through this carefully, methodically, and in practical “what do I actually do with the OCW data” terms, assuming:

New rifle / new year

You already ran a ladder to identify a safe range

Now you want to run an OCW for Charge weight (not SD yet — important distinction)

I’ll be detailed but structured so it stays usable, not overwhelming.

First: clear up one common mix-up

OCW is not primarily for SD.

OCW is for finding a charge weight that is insensitive to small variations — meaning POI stays stable even if velocity wiggles.

Low SD is nice, but:

Stable POI > low SD

You can shoot tiny groups with “crappy ” SD

You can shoot trash groups with “great” SD

Think of SD as a secondary confirmation, not the goal.

Step 1: How OCW data is shot (quick recap)

Load 3 to 5 rounds per charge

Charges are typically 0.2–0.3 gr apart

Same brass, primer, seating depth

Shoot round-robin at separate targets

Distance usually 100 yd (200+ can help but not required)

You end with:

Multiple 3-shot groups

Each group has its own POI

Chrono data (optional but helpful, I almost want to say critical)

Step 2: Ignore group size at first (seriously)

This is where most people go wrong when reviewing OCW data:

■ Don’t chase the smallest group

■ Don’t obsess over SD yet

■ Don’t throw out “ugly” groups immediately

■ Look for POI consistency across adjacent charges

Step 3: Identify the OCW “node”

You’re looking for 2–3 consecutive charge weights where:

Group centers land in nearly the same spot

Vertical dispersion is minimal between them

Horizontal may vary slightly (wind / shooter)

Example:

Charge

POI, 41.6 Low-left

41.8, Center

42.0, Center

42.2, Center

42.4, High-right

■ 41.8–42.2 is your node

Why this matters:

That charge range is forgiving

Temperature, brass volume, or ±0.05gr won't wreck accuracy in fact this is what makes a "match load" reliable

Step 4: Pick the middle of the node

Always choose the middle charge, not the edge.

Why:

Gives you buffer against pressure and offer more stability across temps and is Less sensitive to scale error

From the example:

Node = 41.8–42.2

Pick 42.0

This is now your working charge weight

Step 5: Confirm with 5–10 shot groups

Now you verify, not reinvent.

Load:

Same charge (42.0)

Same seating depth

Shoot one or two 5–10 shot groups

You're checking:

Is vertical stringing minimal?

Is the group repeatable?

Does SD roughly align (don't panic if it's not single-digits)

If groups are trash → revisit OCW

If groups are decent → proceed

Step 6: Only now does SD actually matter

At this stage:

Chrono your confirmed charge

Record ES/SD across 10+ shots

Rules of thumb:

SD ≤ 10 → very good

SD ≤ 15 → perfectly usable

SD > 20 → look at brass prep or primers

Do NOT abandon a good node over SD alone.

Fix SD with prep before changing charge.

Step 7: Seating depth tuning (this is where groups shrink)

Now — and only now — you tune seating depth.

Setup:

Same charge

Same brass lot

Change seating depth only

Typical jump increments: 0.005" or 0.010"

Start at a safe jump (ex: 0.020–0.040" off lands)

Example test: 0.020..0.030..0.040..0.050

Shoot:

3–5 shots per depth

Same target or separate, doesn’t matter

Step 8: Interpret seating depth results

Now you do care about group size.

Look for:

One depth that suddenly tightens

Reduced vertical

Round, predictable groups

Often you’ll see:

One clear winner

Or a small “window” (ex: 0.030–0.040 both shoot well)

Pick: The most forgiving depth, not necessarily the tiniest one-off

Step 9: Final confirmation

Load your “final” recipe:

Charge: OCW middle, seating depth: best performer

Shoot:

One 10-shot group

Or multiple 5-shot groups

If it repeats: ■ You’re done

Big-picture logic (why this works)

OCW gives you charge stability

Seating depth gives you precision

SD tuning gives you consistency at distance

Trying to optimize them out of order:

Wastes components

Chases noise

Creates fragile loads

One-sentence summary

OCW finds a forgiving charge node, you select the center, confirm it, then use seating depth to shrink groups — SD is a supporting metric, not the driver.

SEATING DEPT EXPLAINED Seating Depth Tuning — How Small Groups Are Actually Found

Big Picture First (anchor this mentally)

Powder charge controls barrel timing

Seating depth controls bullet alignment and release

Once your charge is stable (ladder + OCW done), seating depth is what collapses dispersion into a tight cluster.

If you tune seating depth first, you're tuning on shifting ground.

If you tune it last, it works like magic.

Why Seating Depth Matters (the physics)

When a bullet is fired, three critical things happen before it leaves the barrel:

Bullet jump (distance to lands)

Neck tension release

Engraving into rifling

Seating depth directly affects all three.

Small changes = big effects because:

The bullet is transitioning from static → moving → engraved

Any yaw or misalignment here gets locked in

The barrel cannot fix bad entry — only amplify it

The Two Seating Depth Philosophies

1■■ Jump tuning (most modern precision rifles)

Bullet jumps to lands

Very forgiving

Best for mags / temperature stability

2■■ Jam tuning (benchrest / single-feed)

Bullet touches or jams into lands

Extremely sensitive

Very narrow sweet spot

We'll focus on jump tuning, because it's practical and repeatable.

Step 1: Measure the Lands (accurately)

You must know where the lands are for that bullet.

Methods:

OAL gauge (Hornady style)

Modified case

Comparator (not COAL to tip)

■■ Important:

Measure base-to-ogive, not tip

Lands vary by bullet shape

Record:

Base-to-ogive touching lands. This is your zero reference.

Step 2: Choose a Safe Starting Jump

Never start at the lands unless you know the system.

Typical starting jumps:

0.020"..0.030"..0.040"

If unknown: ■ Start at 0.040" off

This avoids:

Pressure spikes

Hard bolt lift

False accuracy due to pressure

Step 3: How Seating Depth Affects Groups

Too close to lands:

Pressure spikes

Vertical stringing

Random flyers

Inconsistent release

Too far from lands:

Bullet enters rifling crooked

Horizontal spread

Larger but “round” groups

Sweet spot:

Clean release

Minimal yaw

Barrel timing + bullet alignment line up

That’s where groups shrink.

Step 4: Designing the Seating Depth Test (properly)

Rules:

One powder charge only (OCW winner)

Same brass lot

Same neck tension

Same primer

Same conditions if possible

Increment size:

- 0.005" for sensitive bullets
- 0.010" for most
- 0.020" if you’re far off lands

Example Test Plan:

Let’s say lands = 2.150" BTO

Test: 2.110 (0.040 off)..2.120 (0.030 off)..2.130 (0.020 off)..2.140 (0.010 off)

Shoot:

3–5 shots per depth

Separate targets preferred

Step 5: How to Read Seating Depth Results (this is critical)

Now — unlike ladder or OCW — group size matters.

Look for:

One depth that is clearly tighter

Reduced vertical AND horizontal

Fewer unexplained flyers

What you’ll often see:

Groups get worse → worse → suddenly tiny → worse again

That tiny one is not random. That’s the alignment window.

Step 6: Understand Seating Depth “Nodes”

Just like powder, seating depth has nodes.

These are:

Ranges where bullet entry is consistent. Not always a single number

Example:

0.030 = 0.6 MOA

0.025 = 0.35 MOA

0.020 = 0.32 MOA

0.015 = 0.65 MOA

■ Node = 0.020–0.025

Pick the middle, not the extreme.

Why?

Throat erosion

Minor seating variation

Temperature stability

Step 7: Fine Tuning Inside the Node (optional but deadly)

Once you find a promising window:

Narrow increments (0.003–0.005")

Example: 0.018..0.020..0.022

Shoot 5-shot groups.

This is how sub-¼ MOA loads are found.

Step 8: Confirm With Volume (truth test)

Never trust one tiny group.

Load:

Final seating depth

Same charge

Shoot:

One 10-shot group or three 5-shot groups

If it repeats: ■ That’s your load.

Seating Depth vs SD (important reality check)

Seating depth:

Rarely improves SD

Often slightly worsens it

Dramatically improves grouping

That’s normal.

Accuracy ≠ low SD.

Why Seating Depth Is So Powerful

Powder tuning:

Adjusts when the bullet exits

Seating depth:

Adjusts how the bullet enters the barrel

Bad entry = no amount of powder fixes it

Good entry = average velocity shoots tiny

Common Seating Depth Mistakes

Changing charge and depth at same time

Measuring COAL to tip

Testing too close to lands first

Declaring victory after 3 shots

Chasing the smallest single hole instead of repeatability

Throat Erosion Reality (long-term accuracy)

As you shoot:

Lands move forward

Jump increases

Groups slowly open

Fix:

Periodically chase your seating depth

Maintain the same jump, not same COAL

This is how match shooters keep accuracy for 1–2k rounds.

Final Mental Model (lock this in)

Charge weight = stability

Seating depth = alignment

Alignment = group size

One-sentence summary:

Seating depth tuning works by finding the jump where bullet release and rifling entry are most consistent, collapsing dispersion into tight, repeatable groups once charge stability is already solved.